

Semantic Networks:

Mapping relationships between Words,
Documents, Sentences, and Topics

Large Language Models, Text Embeddings,
and Network Analysis

Sunbelt 2026, Afternoon Session



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Where are the networks? Everywhere.

- Many NLP techniques—topic modelling especially—include some form of relational component (even if you disregard semantic similarity).
- The document-term matrix I introduced with the work of Gerard Salton is ostensibly a bipartite network of documents and words.
 - These document-term matrices were the foundation of NLP for decades.
- The sliding window employed by word2vec can also be used to map word cooccurrence networks.
 - TextRank extracts keywords from text by doing exactly this!

(Mihalcea & Tarau, 2004)

Where are the networks? Everywhere.

- Some authors have directly applied topic modelling techniques to social network data, such is their structural similarity. (Henderson & Eliassi-Rad, 2009; Cha & Cho, 2012)
- The output of early topic models—latent semantic analysis (LSA) and latent Dirichlet allocation (LDA)—is a pair of weighted bipartite networks that connect words/documents to topics. (Deerwester et al., 1990; Blei et al., 2003)
- Blei and Lafferty (2006) formally integrated networks into LDA with the correlated topic model

Where are the networks? Everywhere.

- Two popular embedding-based topic models from the last 6-years—top2vec and BERTopic—integrate a method called “Uniform Manifold Approximation and Projection” (UMAP).
- UMAP is a method designed to take high-dimensional vectors (i.e., embeddings) and flatten them into fewer dimensions, while retaining information.
- Achieved by converting the high-dimensional embeddings into a network, and then finding low-dimensional embeddings that replicate that network.

(McInnes et al., 2018; Angelov, 2020; Grootendorst, 2022)

Where are the networks? Everywhere.

- Multitudes of researchers have published a variety of novel integrations NLP and network science methods; beyond those I have mentioned.
 - Networks can be used to refine or expand on results from topic models.
 - Topic models can construct or augment data for network analysis.
 - Networks can be an integral component in the estimation of NLP methods.
(see Smith & Vacca, 2025 for review)
- Suffice it to say, teaching you every way that NLP and network science intersect would take an entire semester!

Embeddings → Networks

- For the purposes of this workshop, we are going to focus on the most recent (and arguably robust) approach to building **semantic** networks based on embeddings.
 - These networks will map the relationships between one or more of:
 - Words
 - Sentences
 - Documents
 - Topics
- These are your nodes!*

Embeddings → Networks

- There are a multitude of ways you might approach building a network from a cloud of points (i.e., embeddings).
- For this workshop, we are focusing on three approaches:
 - Naïve thresholding semantic similarity matrices.
 - ‘Sparsifying’ semantic similarity matrices with **backbone extraction**.
 - **Filtration** of the text embeddings vis-à-vis topological data analysis.
- These approaches both aim to identify which nodes (words, sentences, documents, or topics) are sufficiently proximal to merit being ‘connected’.

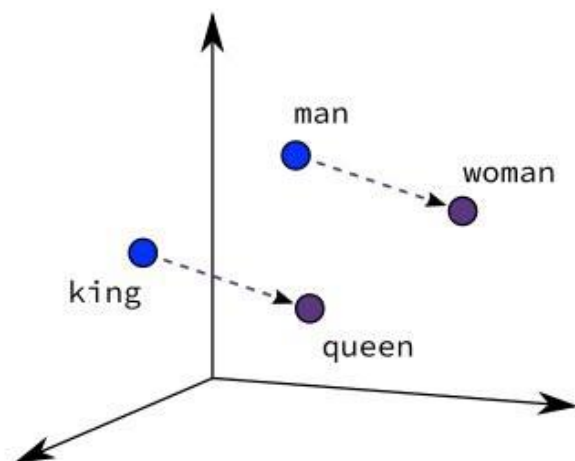
Semantic Similarity

- Our goal, as network scientists, is to examine relationships.
- Once we have numeric representations of our words, documents, sentences, topics, etc. it is *very* easy to extract relational information.
- Indices of the relationship between text embeddings are collectively called 'semantic similarity' measures.
- However, keep in mind that this terminology arose in computer science, not linguistics.

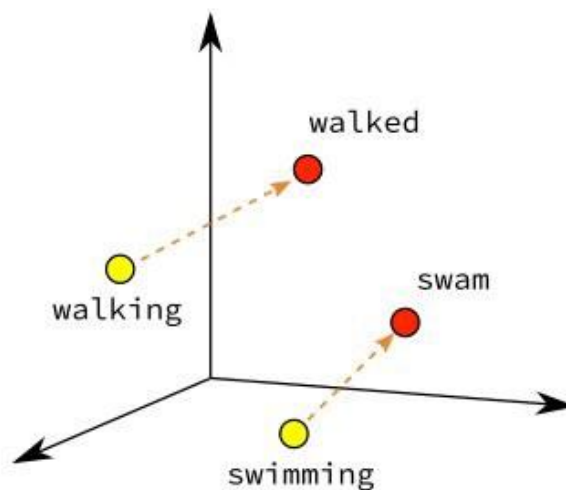
Semantic Similarity

- However, keep in mind that this terminology arose in computer science, not linguistics.
- Natural language processing methods operate across a variety of different linguistic levels at all stages of the process (input, training, and output).
- This means the interpretation of 'semantic similarity' is variable.
 - E.g., Word2Vec's tendency to capture semantic v. syntactic meaning.

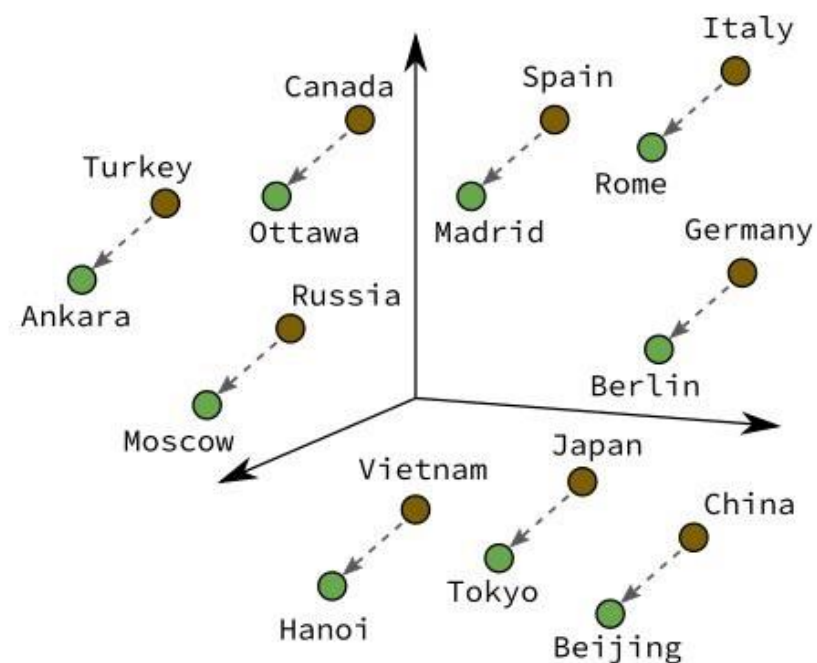
Semantic Similarity



Male-Female



Verb Tense



Country-Capital

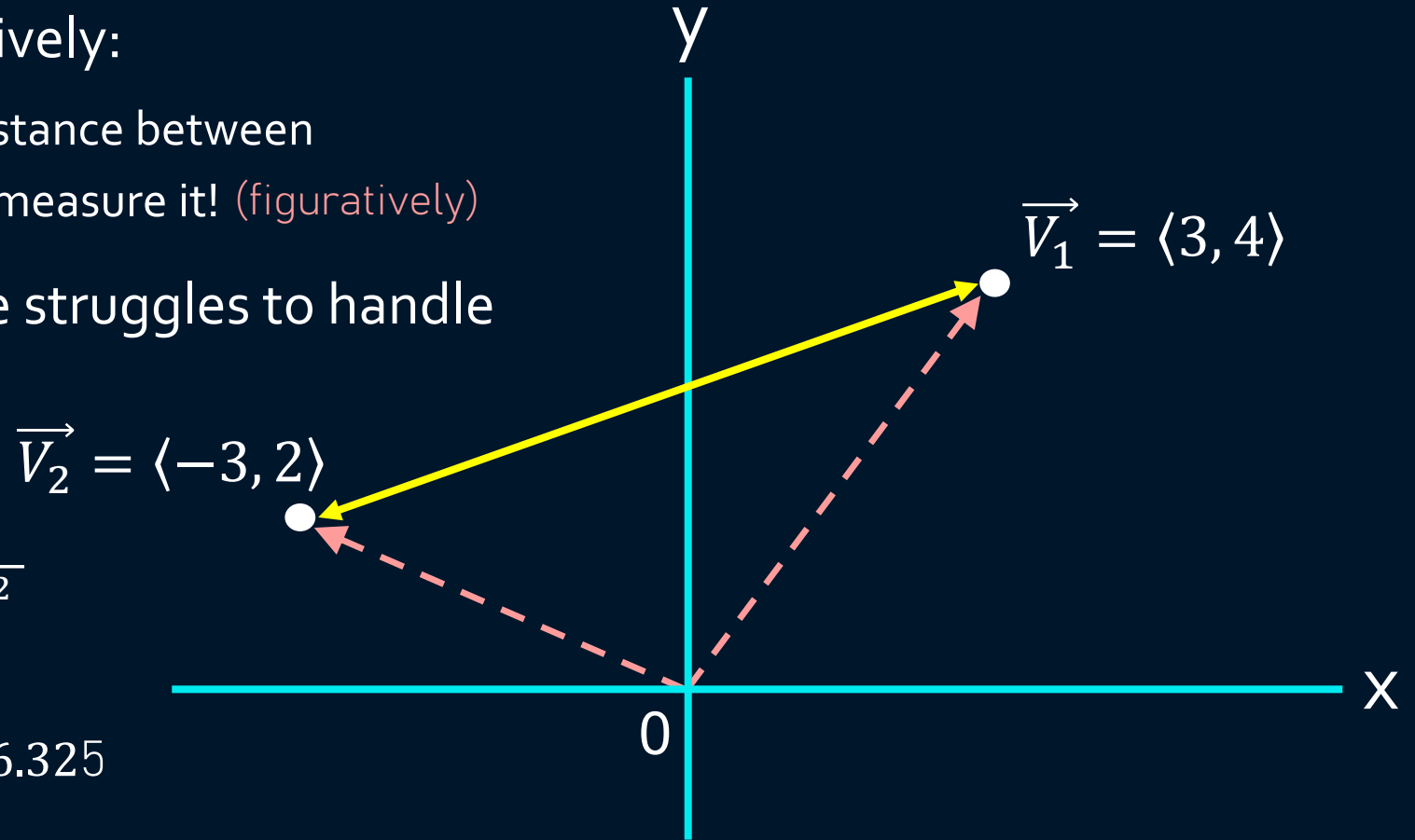
<https://developers.google.com/machine-learning/crash-course/embeddings/translating-to-a-lower-dimensional-space>

Euclidean Distance

- Makes the most sense intuitively:
 - If you want to determine the distance between two points, pull out a ruler and measure it! (figuratively)
- However, Euclidean distance struggles to handle high dimensional space.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{((-3) - 3)^2 + (2 - 4)^2} = 6.325$$



Dot Product

- The most difficult measure to intuit—so I am going to wiggle the second vector around a bit.
- It generally represents the extent that two vectors point in the same direction.
- Visualizing a projected shadow helps understand the intuition of the measure.

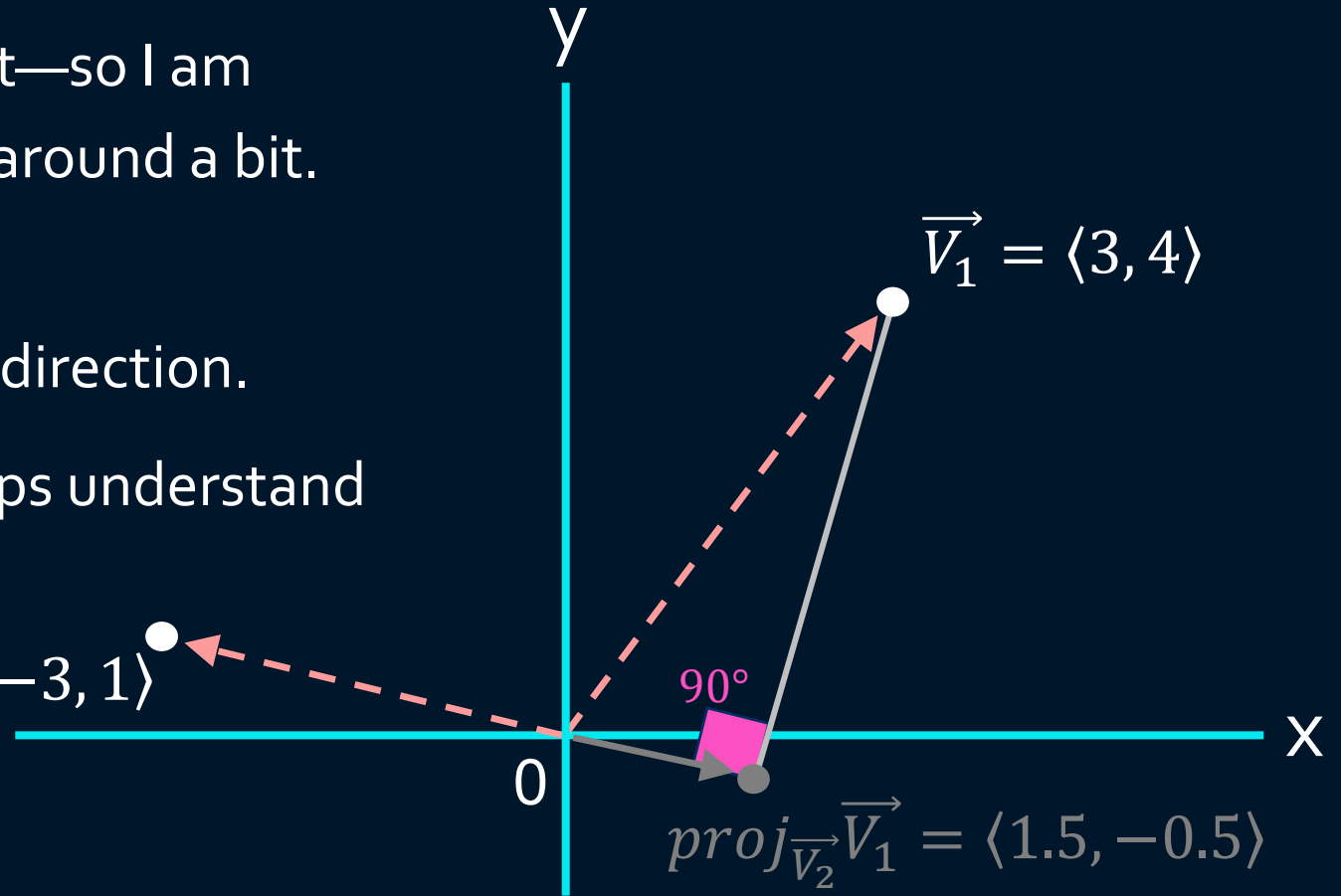
$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = (3)(-3) + (4)(1) = -5$$

$$\vec{V}_2 = \langle -3, 1 \rangle$$

$$\vec{V}_1 = \langle 3, 4 \rangle$$

$$\text{proj}_{\vec{V}_2} \vec{V}_1 = \langle 1.5, -0.5 \rangle$$

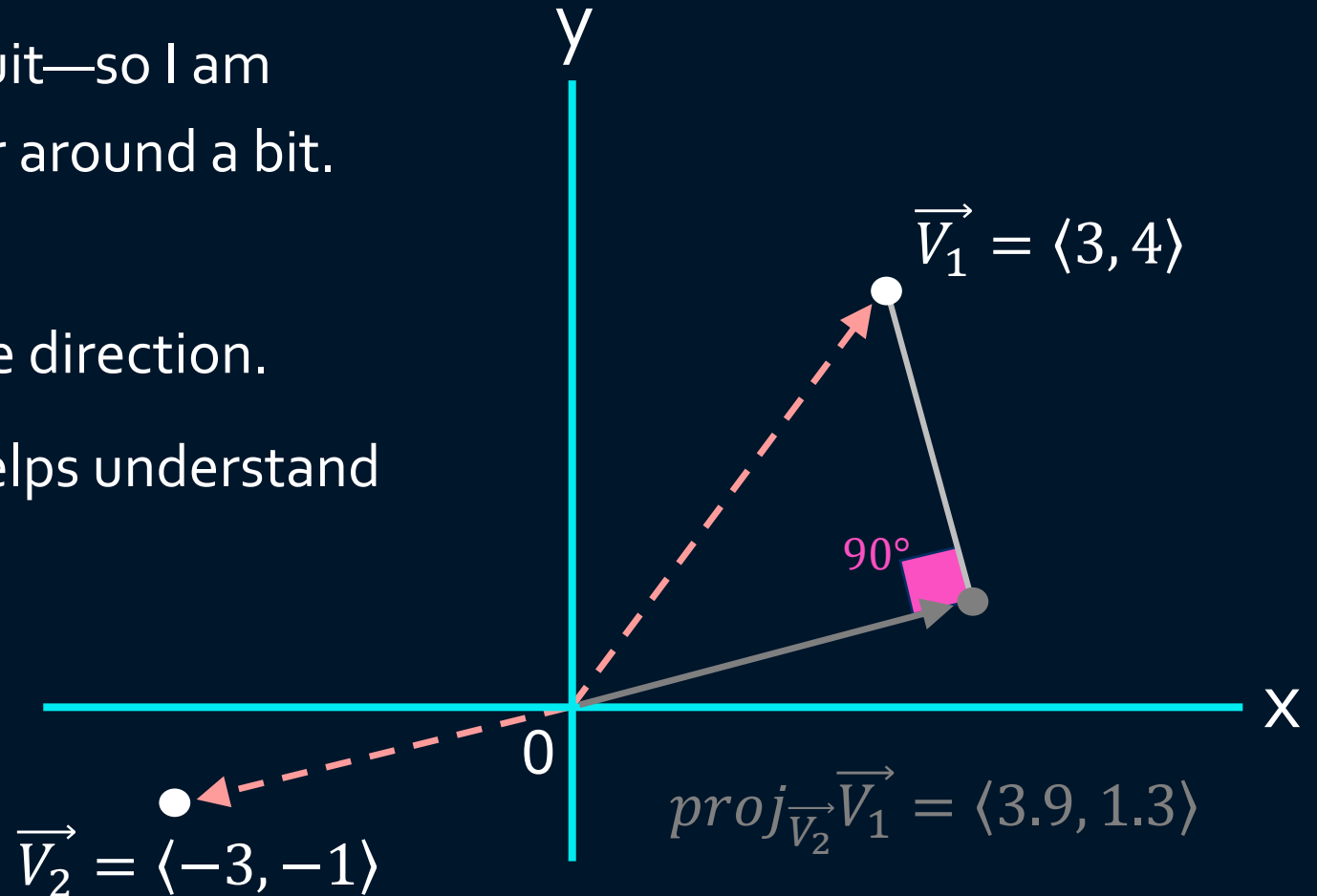


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$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = (3)(-3) + (4)(-1) = -13$$

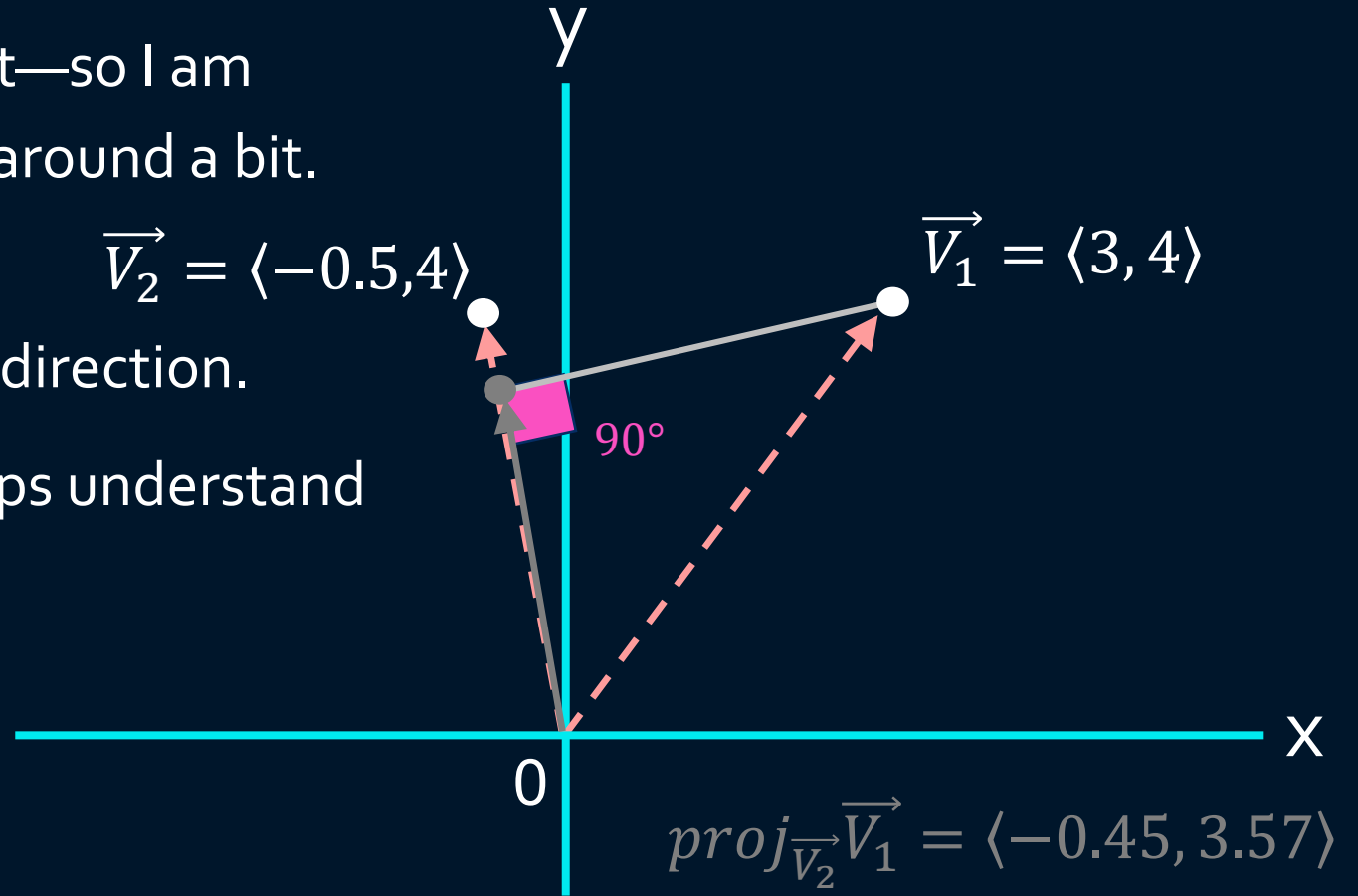


Dot Product

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$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = (3)(-0.5) + (4)(4) = 14.5$$



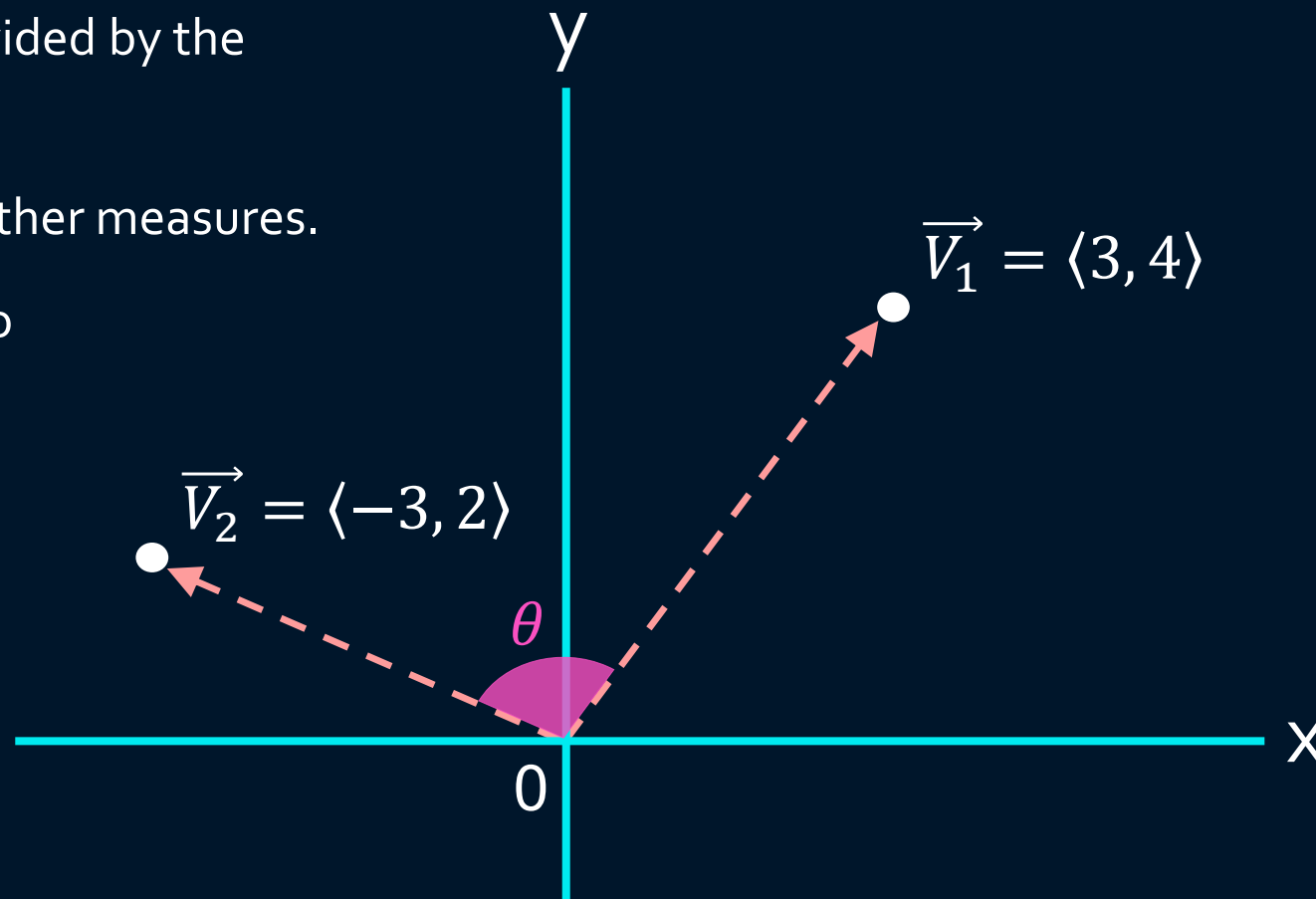
Cosine Similarity

- It is the dot product of the two vectors divided by the product of their magnitudes (lengths).
- Scales into high dimensions better than other measures.
- This is the most common measure used to quantify semantic similarity.

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| ||\vec{b}||}$$

$$||\vec{a}|| = \sqrt{a_1^2 + a_2^2 + \dots + a_k^2}$$

$$||\vec{b}|| = \sqrt{a_1^2 + a_2^2 + \dots + a_k^2}$$



Cosine Similarity

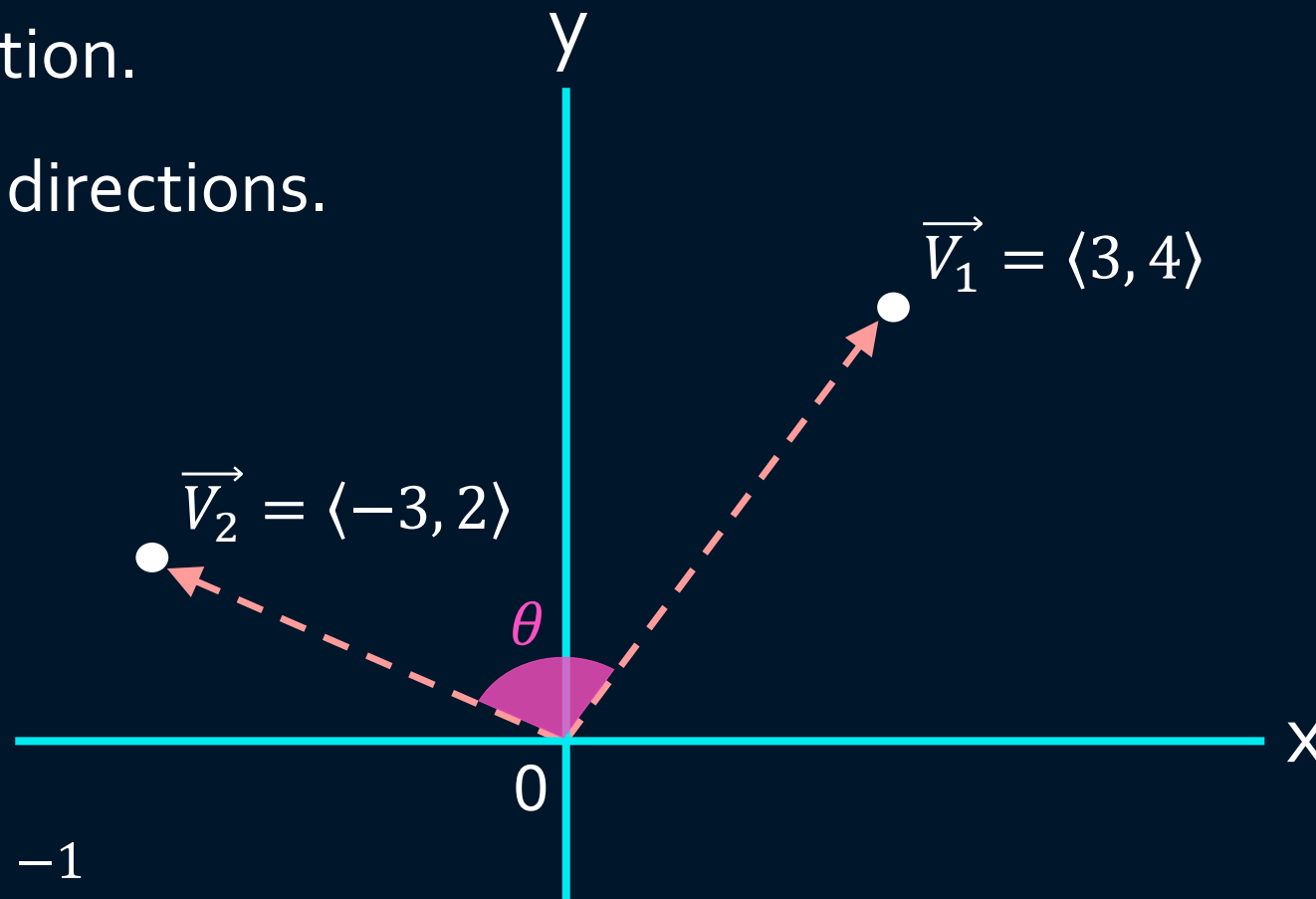
- $\cos\theta \rightarrow 1$ denotes same direction.
- $\cos\theta \rightarrow -1$ denotes opposite directions.

$$\cos\theta = \frac{-1}{5 \cdot 3.606} \approx -0.055$$

$$||\vec{a}|| = \sqrt{3^2 + 4^2} = 5$$

$$||\vec{b}|| = \sqrt{(-3)^2 + 2^2} \approx 3.606$$

$$\vec{a} \cdot \vec{b} = (3 \times (-3)) + (4 \times 2) = -9 + 8 = -1$$



Cosine Similarity

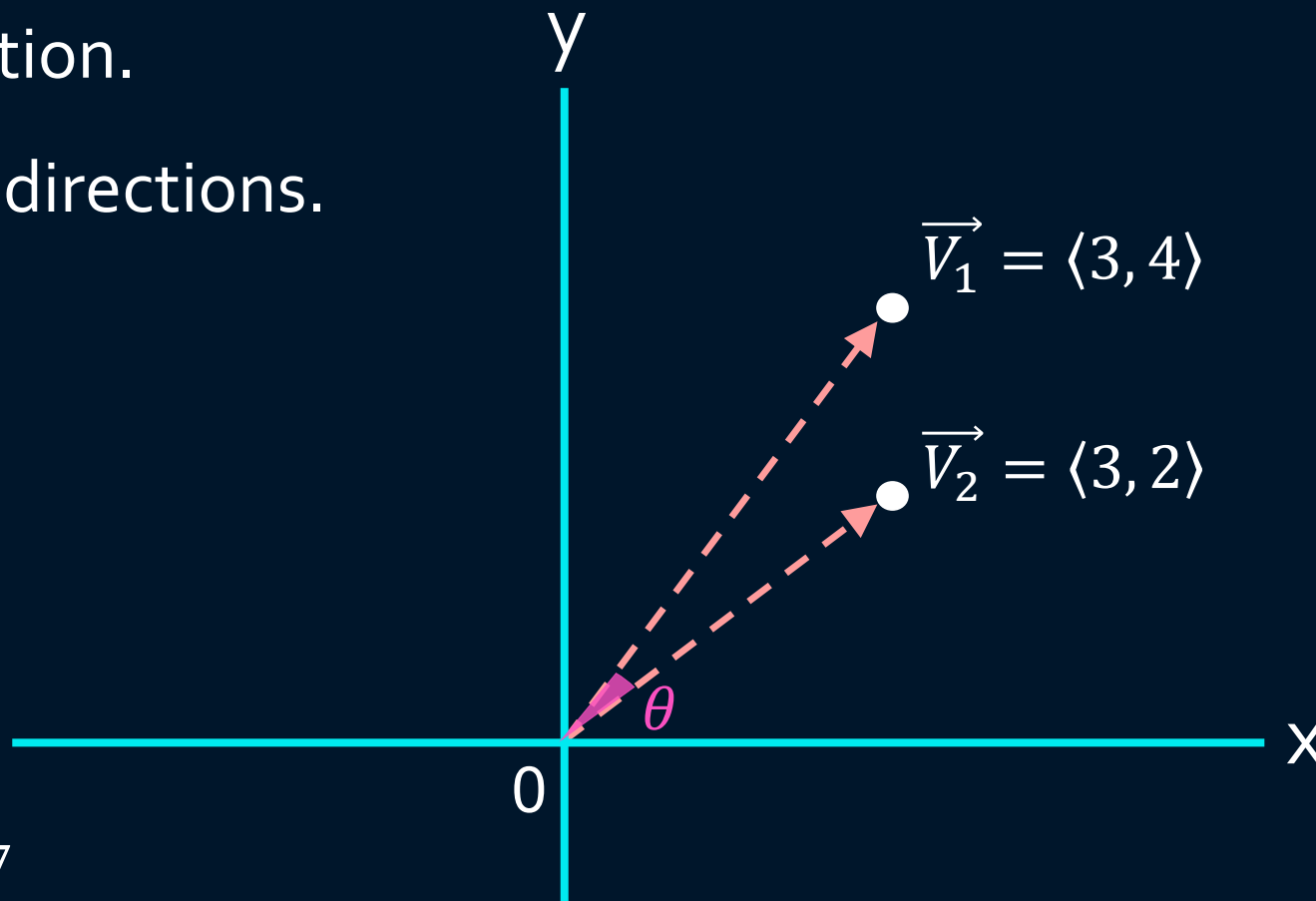
- $\cos\theta \rightarrow 1$ denotes same direction.
- $\cos\theta \rightarrow -1$ denotes opposite directions.

$$\cos\theta = \frac{17}{5 \cdot 3.606} \approx 0.943$$

$$||\vec{a}|| = \sqrt{3^2 + 4^2} = 5$$

$$||\vec{b}|| = \sqrt{3^2 + 2^2} \approx 3.606$$

$$\vec{a} \cdot \vec{b} = (3 \times 3) + (4 \times 2) = 9 + 8 = 17$$



Cosine Similarity

- Semantic similarity measure akin to correlation.
- Assesses similarity between two vectors (variables) based on a tendency to trend in the same direction.
- Like correlation, the output is an **adjacency matrix**.
- Serves as starting point, because it can be treated as a **maximally connected, undirected, weighted (and usually signed) network**.

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| ||\vec{b}||}$$

↓

	W_1	W_2	W_3	W_4	W_5
W_1	1				
W_2	.63	1			
W_3	.85	.74	1		
W_4	-.21	-.11	-.09	1	
W_5	-.35	.01	-.01	.93	1

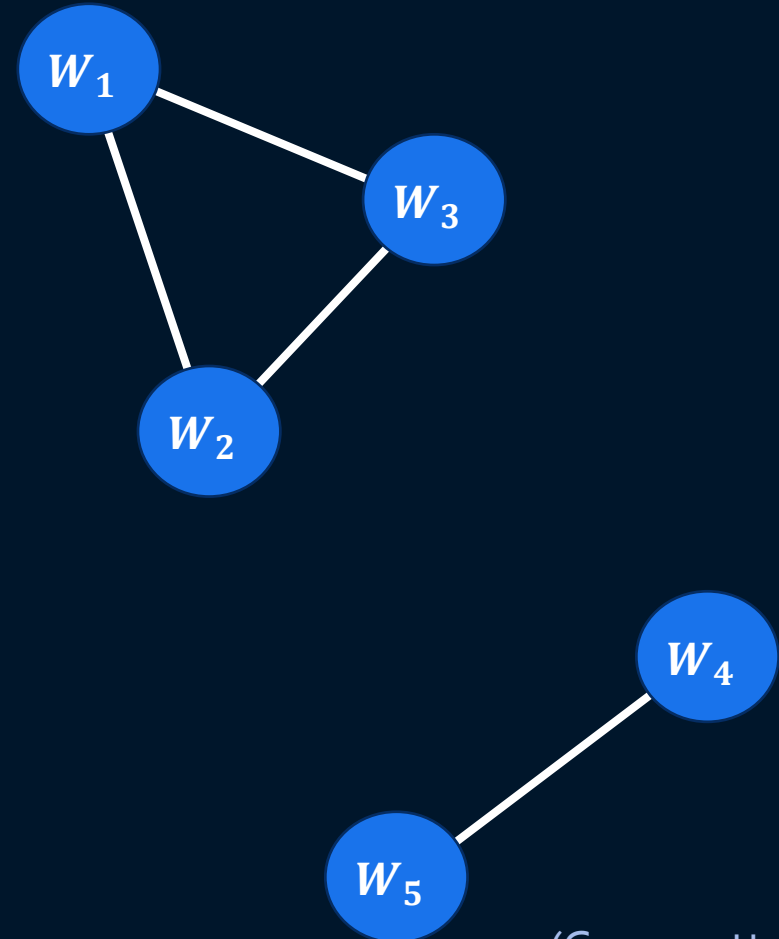
(Naïve) Global Thresholding

- Starting with the example semantic similarity matrix from the previous slide, we can quickly see clusters of related words:
 - Cluster A:** $\{W_1, W_2, W_3\}$
 - Cluster B:** $\{W_4, W_5\}$
- The similarity metric for these two clusters are all above 0.5, so you could simply impose a threshold of 0.5.

	W_1	W_2	W_3	W_4	W_5
W_1	1				
W_2	.63	1			
W_3	.85	.74	1		
W_4	-.21	-.11	.02	1	
W_5	-.35	.25	.01	.93	1

(Naïve) Global Thresholding

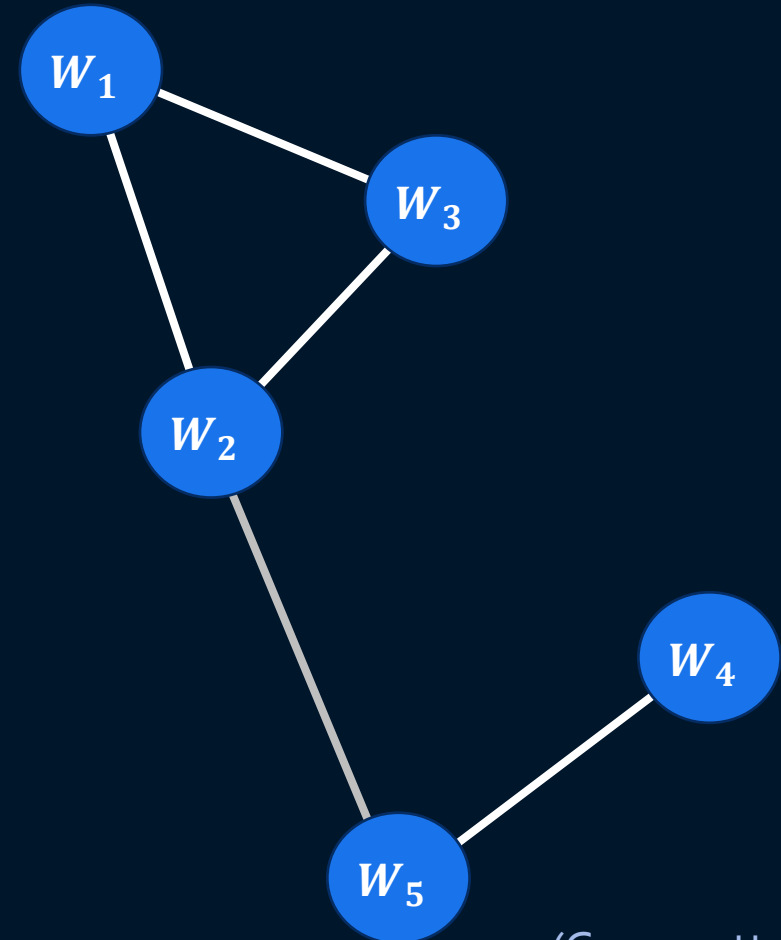
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(Granovetter, 1978)

(Naïve) Global Thresholding

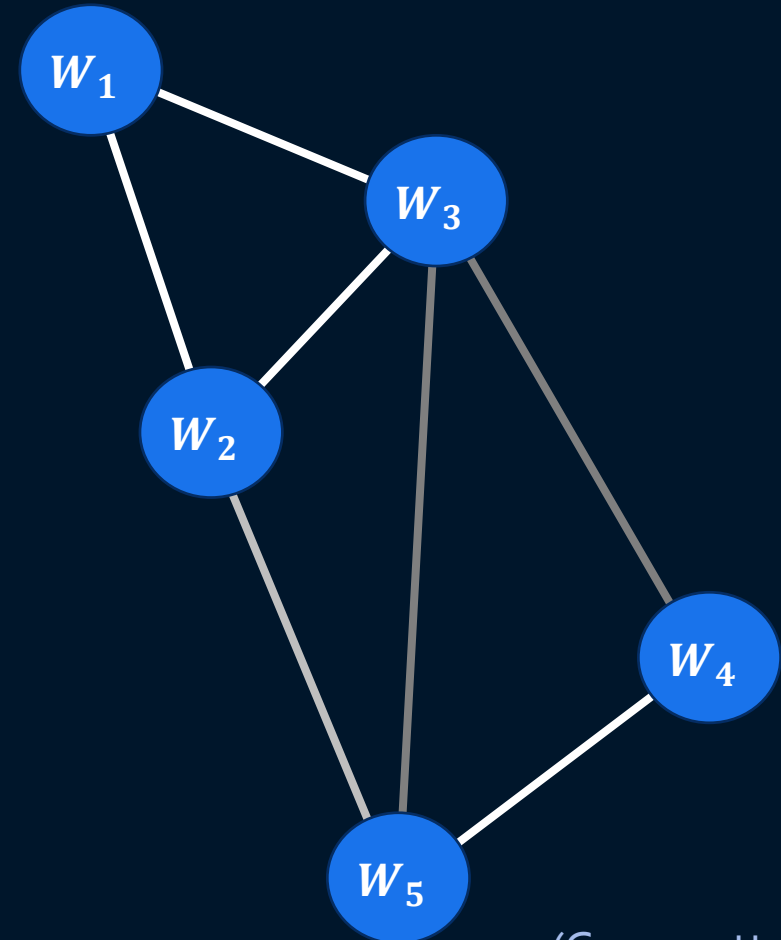
- If this network is not to your taste (maybe it is too fragmentary), you could increase the threshold.
- Now we have a threshold of 0.25.
- More of a structure is emerging.



(Granovetter, 1978)

(Naïve) Global Thresholding

- Alternatively, we could set the threshold to 0 and allow any positive relationship between words.
- Now we have a network with a meaningful structure rather than a pair of maximally connected components.



(Granovetter, 1978)

(Naïve) Global Thresholding

- Two related problems emerge:
 1. What arbitrary threshold(s) do you impose?
 2. Semantic similarity distributions are dependent on dimensionality and training (and tend to be inconveniently distributed).
- Low dimensional → uniform distribution.
- High dimensional → normal distribution.
- In either case, there isn't an easy-to-identify point where words become meaningfully related.

Note: There *are* many structural backbone extraction methods that will focus more on the global structure of the cosine similarity matrix, but—theoretically—it makes more sense to focus on the local relationships between words, rather than the global.

There is evidence that semantic networks are prone to preferential attachment, meaning well-connected words accrue connections to other words simply by virtue of their connectivity.

Rich words get richer.

(Steyvers & Tenenbaum, 2005;
Mak & Twitchell, 2020)

Backbone Extraction

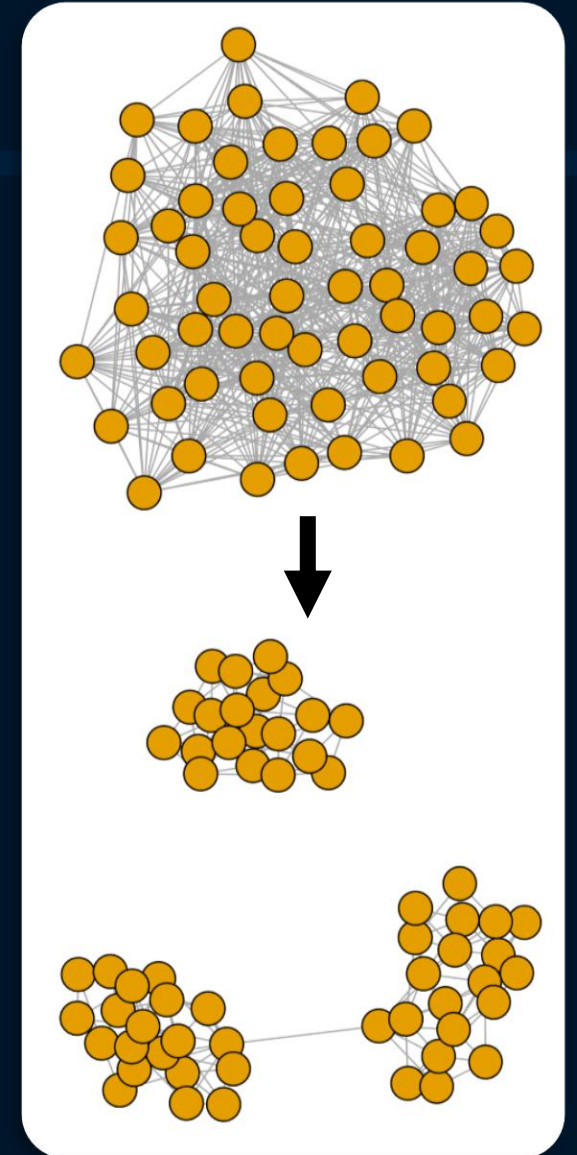
- **Goal:** find a method that *empirically* finds *unusually* high semantic similarities.
- To achieve this goal, we can turn to the backbone extraction methods proposed by Serrano (2009), Foti et al. (2011), and others.
 - *If you need a more robust introduction to these methods, reach out to Zachary Neal—he's the expert!*
- We are going to focus on LANS—Locally Adaptive Network Sparsification.



LANs: Locally Adaptive Network Sparsification

- A non-parametric backbone extraction technique.
- Intended to handle preferential attachment.
 - LANs calculates the statistical significance of edges based on a local empirical cumulative density function (ECDF).
- This benefits semantic networks, as it allows words that are semantically isolated (weaker-than-average cosine similarities) retain locally important connections.
- Additionally, it prunes the weaker ties around weighted degree central nodes in the network.
 - Important, considering we start out maximally connected!

(Foti et al., 2011)



(Neal, 2026)

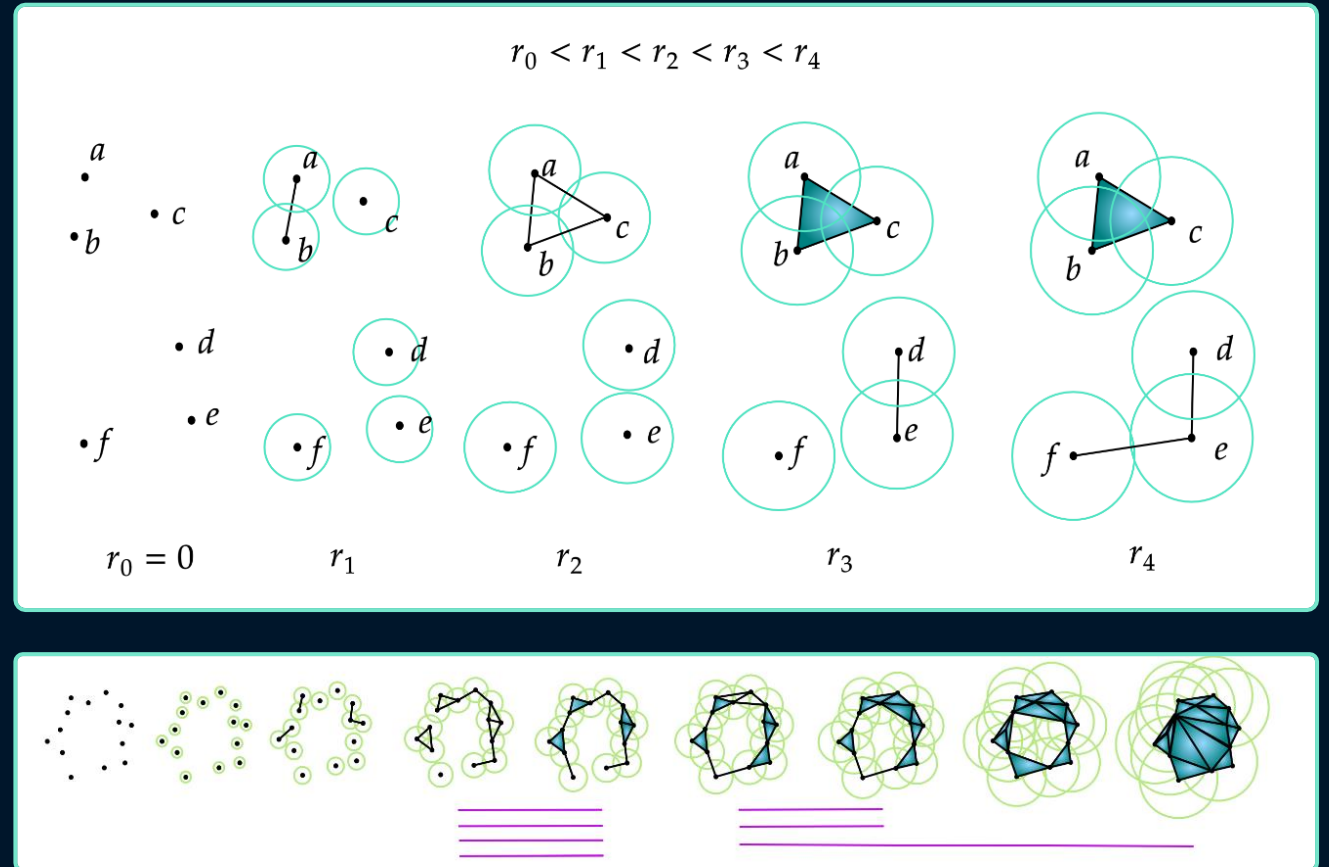
Other Statistical Backbone Extraction Methods

- Disparity Filter
 - Compares observed distribution of edge weights incident on a given node follows a uniform distribution; determines significance based on that assumption (parametric).
- Marginal Likelihood Filter
 - Functions edge-wise rather than node-wise, treats integer-weighted edges as multiple edges in a binomial distribution, from which significance is determined (parametric).
- Noise Corrected Filter
 - Binomial but uses a Bayesian framework to generate posterior variances for the edges in the network, only removing edges if the weight of that edge sufficiently deviates from the expected value (parametric).
- Enhanced Configuration Model Filter
 - Extension of the marginal likelihood filter but generates a more appropriate null model with the same degree and strength distribution as the actual network based on maximum entropy (parametric).
- And many, *many* more...

(see Yassin et al., 2025 for review and comparison)

Vietoris-Rips and Čech Filtration

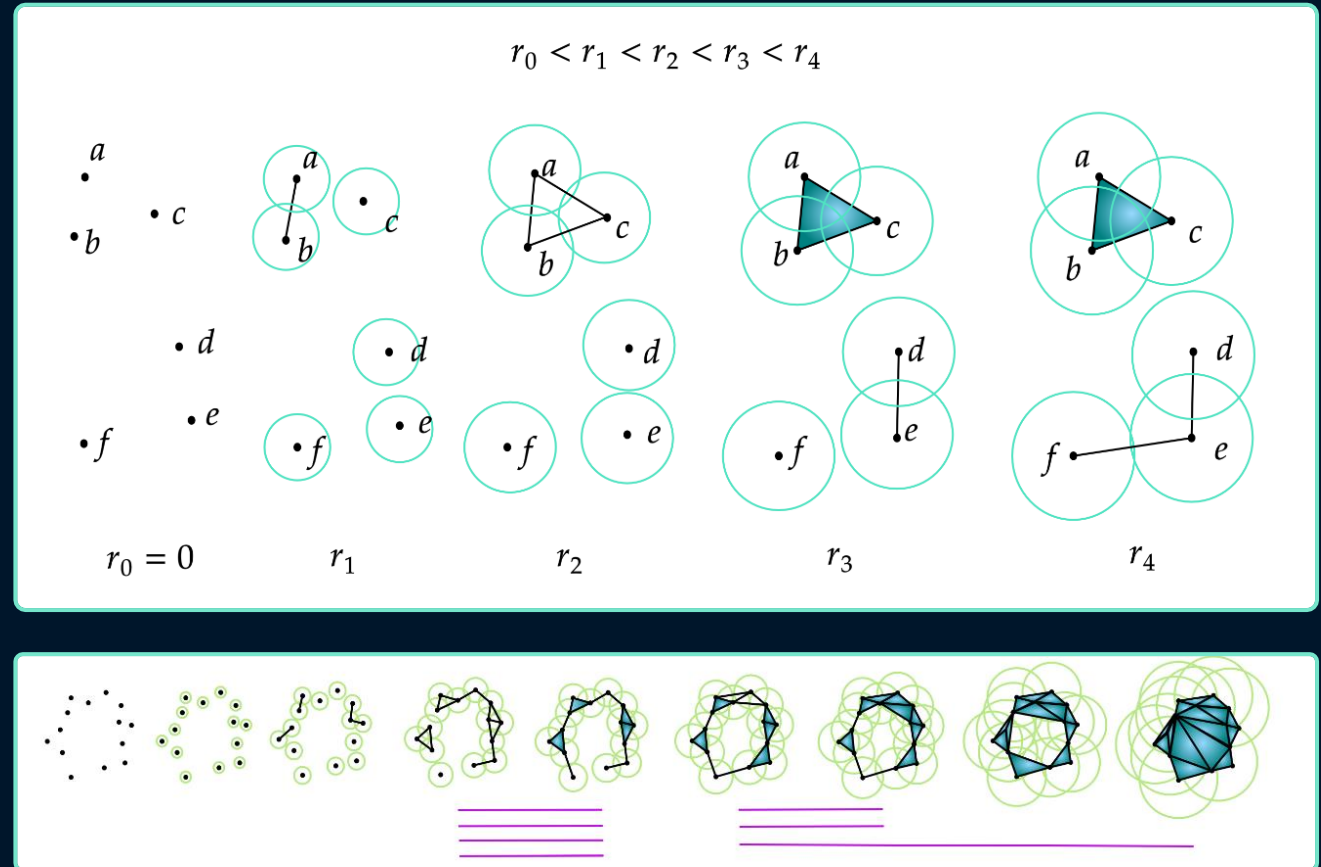
- An important component of **topological data analysis**, filtration techniques are established approaches to building simplicial complexes (i.e., networks) based on points in high-dimensional space.



(Yadav et al., 2025)

Vietoris-Rips and Čech Filtration

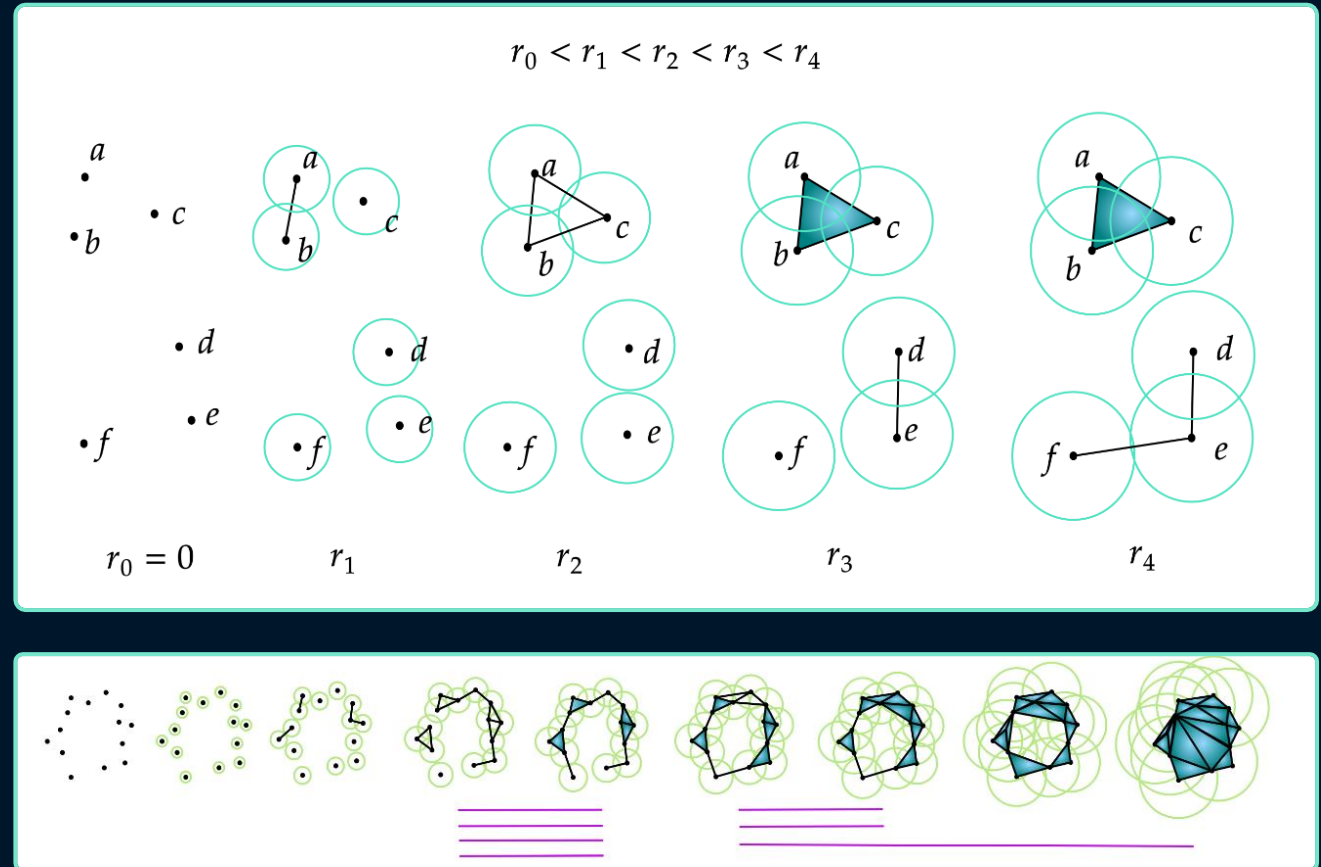
- These techniques are typically employed to analyze 'birth' and 'death' of topological features with the addition of more edges to a network.
- However, we are only interested in the techniques employed that take us from *point cloud* to *network*.



(Yadav et al., 2025)

Vietoris-Rips and Čech Filtration

- These **filtration** methods build 'simplicial complexes' (networks!) by:
 - Placing a cover (circle or ball) over each point in a cloud.
 - Incrementally increasing the radius r of the cover until the covers overlap.
 - Introducing an edge whenever covers start to overlap.



(Yadav et al., 2025)

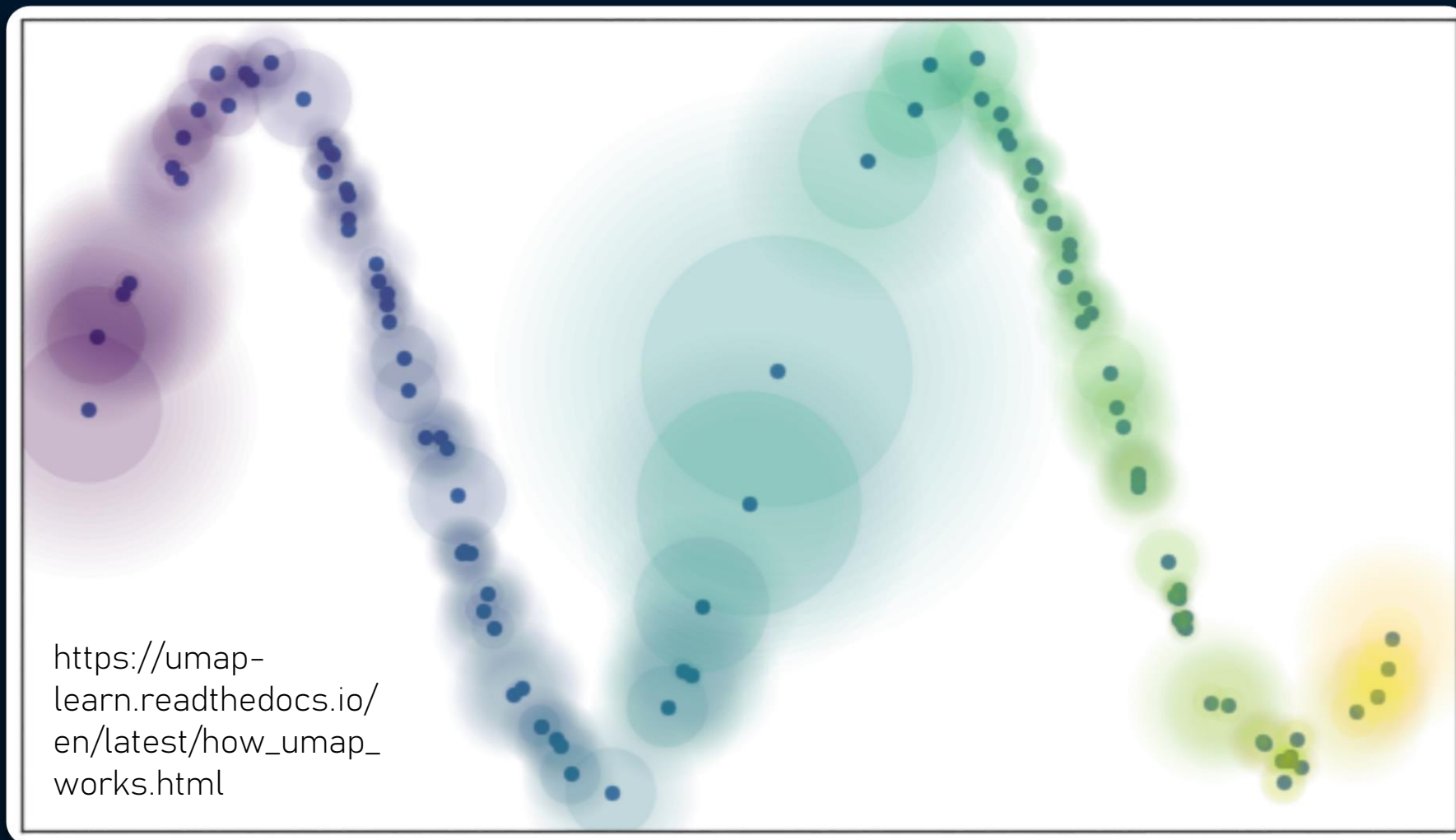
Čech Complexes, UMAP, and Topic Embedding Models

- Top2vec and BERTopic *both* utilize UMAP during estimation, wherein they build semantic networks in order to reduce the dimensionality of text embeddings.
- UMAP incidentally presents a very helpful procedure for building a *fuzzy* network (simplicial complex).
 - Drawing an edge between points in a cloud no longer treated as a binary decision.
 - The covers (circles, balls) atop the points in the cloud are extended variably based on proximity to the first nearest neighbor.
 - The weight of the connection to neighbors beyond the first decays with distance.
 - This approach is arguably the most robust because it corrects for dense clusters, prunes extremely distal edges, and integrates a decaying distance measure as an edge weight.

Simplicial (Čech) Complexes and UMAP

- UMAP building a simplicial (Čech) Complex using open covers.

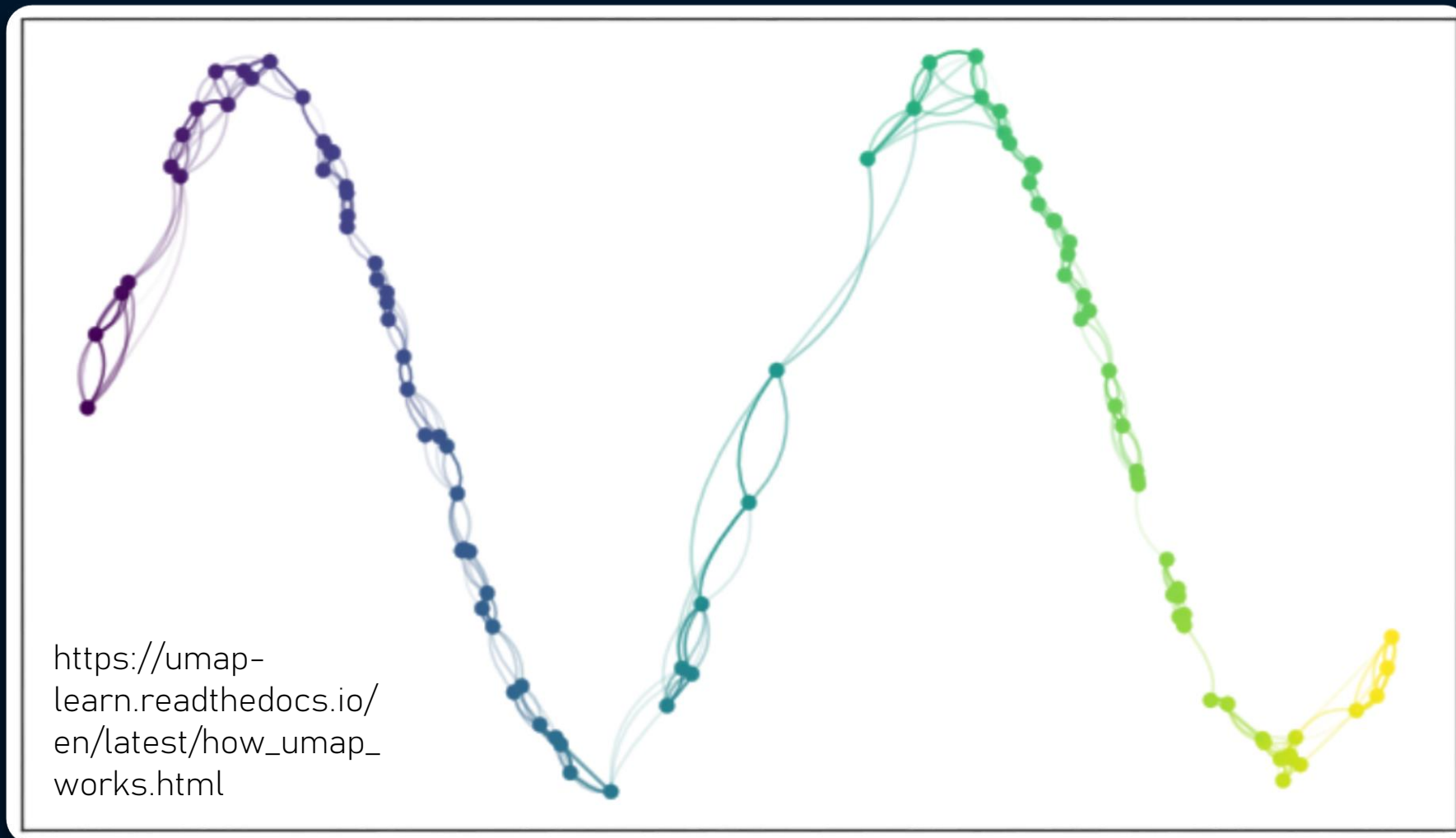
(McInnes et al., 2018)



Simplicial (Čech) Complexes and UMAP

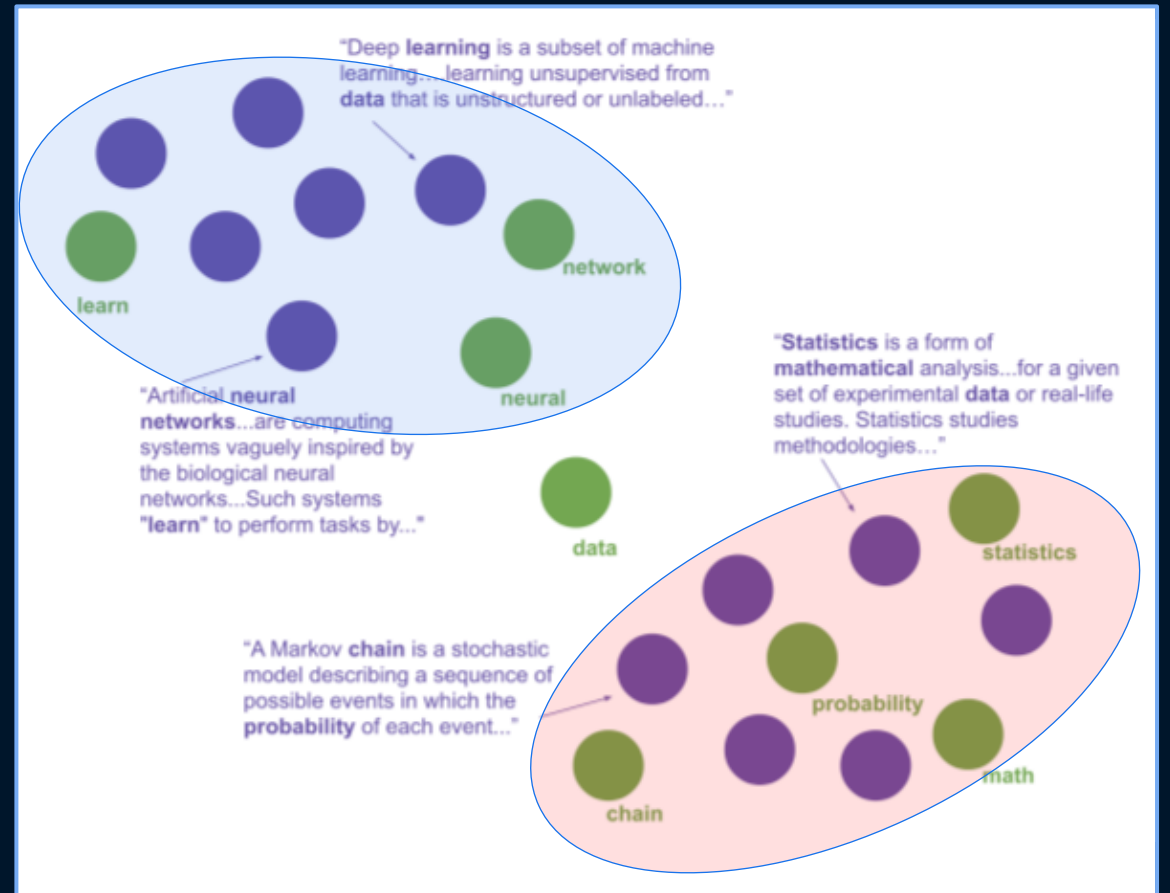
- UMAP
building a
simplicial
(Čech)
Complex
using open
covers.

(McInnes et al., 2018)



Top2Vec

- Word embeddings generated using the skip-gram word2vec model.
- Interleaved doc2vec embeddings using the distributed bag-of-words (DBOW) model.
- Reduce dimensionality using *uniform manifold approximation and projection for dimension reduction (UMAP)*.
- Use hierarchical density-based spatial clustering of applications with noise (HDBSCAN) to find clusters of documents and words – aka topics.



(Angelov 2020)

BERTopic

- Functions largely the same way as top2vec, but starts with a pre-trained, compact large language model (all-MiniLM-L6-v2 is the default):
 1. Vectorization (generate document embeddings based on the seed model).
 2. Dimensionality reduction (UMAP).
 3. Clustering (HDBSCAN).
 4. Topic Representation (identifies words that are unique to each cluster using class-based TF-IDF).
- 1. Both models are used to find **topic embeddings**, providing you with a means to simplify your semantic network and focus on the relationships between *themes*.

Topic Networks

- Given that the topics identified by top2vec and BERTopic are represented by topic embeddings, all of the lessons from earlier in this workshop still apply!
- Only now you are mapping the semantic overlap between themes.



Topic 1

Topic 2

Topic 3

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